WORKED SOLUTION

STEP III Specimen paper, Question 11

Mechanics • First-Order Differential Equations

STEP III Specimen paper, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 11.13, *The Snowplough Problem*: https://beyondstep.keithmcnulty.org/ch11/snowplough.html

The key idea

The speed of each plough is given in terms of the snow depth, and the depth depends on *when* the plough reaches a point. So it is natural to treat time as a function of position and write down $\mathrm{d}t/\mathrm{d}x$, the reciprocal of the speed. For the first plough the depth is just the original depth plus what has fallen; for the second, the key observation is that the snow in front of it has only been falling since the first plough cleared that spot.

Solution

Model each plough as a point moving along the road, and measure distance from the common starting point. Throughout, depths are in metres, times in seconds and distances in metres.

Part (i)

At time t, the snow that the first plough meets has been falling undisturbed since $t = 0$, so its depth is

$$z = h + kt.$$

The speed is therefore

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{k}{\alpha(h+kt)}, \qquad \text{so} \qquad \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{\alpha(h+kt)}{k} = \alpha t + \frac{\alpha h}{k}.$$

(Inverting is fine because the speed is positive, so x is a strictly increasing function of t.) This is linear in t, and it is neatest to write it as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(t + \frac{h}{k}\right) = \alpha\left(t + \frac{h}{k}\right).$$

Hence $t + h/k = Ce^{\alpha x}$ for a constant C. At $x = 0$ we have $t = 0$, so $C = h/k$, and the time taken to travel x metres is

$$t_1(x) = \left(e^{\alpha x} - 1\right)\frac{h}{k} \text{ seconds},$$

as required.

Part (ii)

Setting up the equation. Suppose that at time $t > T$ the second plough is at position y. The first plough passed this point at time $t_1(y)$ from part (i) and cleared all the snow there.

Ask yourself What is the depth of snow in front of the second plough?

Not $h + kt$: the first plough left bare road behind it. At position y the snow has been building up from zero since time $t_1(y)$, at rate k, so the depth is

$$z = k\big(t - t_1(y)\big).$$

(This needs $t > t_1(y)$, i.e. the second plough is still behind the first, which is true until they meet.) The speed of the second plough is therefore

$$\frac{dy}{dt} = \frac{k}{\alpha k\big(t - t_1(y)\big)} \qquad \Longrightarrow \qquad \frac{1}{\alpha}\frac{dt}{dy} = t - \left(e^{\alpha y} - 1\right)\frac{h}{k},$$

which is the given equation. The second plough starts from the origin at time T, so the initial condition is $t = T$ when $y = 0$.

Verifying the solution. Write the proposed solution as

$$t = t_1(y) + g(y), \qquad g(y) = \left(T - \frac{\alpha h y}{k}\right)e^{\alpha y}.$$

Then $t - t_1(y) = g(y)$, so we need $\frac{1}{\alpha}\,dt/dy = g(y)$. Differentiating, with the product rule for g,

$$\begin{aligned}\frac{dt}{dy} &= \frac{\alpha h}{k}e^{\alpha y} - \frac{\alpha h}{k}e^{\alpha y} + \alpha\left(T - \frac{\alpha h y}{k}\right)e^{\alpha y} \\ &= \alpha\, g(y),\end{aligned}$$

where the first term is $t_1'(y)$ and the next two are $g'(y)$. So the differential equation holds. Also, at $y = 0$, $t = 0 + T \cdot 1 = T$, so the initial condition holds too.

Why is this *the* required solution? The equation is linear and first order: if t and $\tilde{t}$ both satisfy it, their difference u satisfies $du/dy = \alpha u$, so $u = u(0)e^{\alpha y}$. Both satisfy $t(0) = T$, so $u(0) = 0$ and $u \equiv 0$. Hence there is exactly one solution through $(y, t) = (0, T)$, and we have found it.

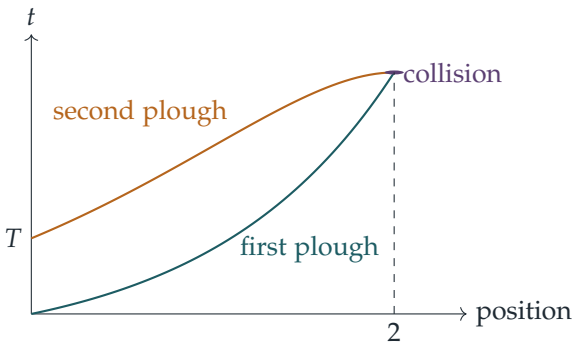
(If you prefer to derive it: multiplying by the integrating factor $e^{-\alpha y}$ gives $\frac{d}{dy}\left(te^{-\alpha y}\right) = -\frac{\alpha h}{k}\left(1 - e^{-\alpha y}\right)$, and integrating from 0 to y leads to the same formula.)

The collision. The ploughs collide at a point y if they reach it at the same time, that is, if $t = t_1(y)$, which means $g(y) = 0$. Since $e^{\alpha y} > 0$, this happens exactly when

$$T - \frac{\alpha h y}{k} = 0 \qquad \Longleftrightarrow \qquad y = \frac{kT}{\alpha h}.$$

For $0 \leq y < kT/(\alpha h)$ we have $g(y) > 0$ (taking $\alpha, h, k > 0$, as the speeds must be positive), so the second plough reaches each such point strictly after the first and they do not meet earlier. Hence the snowploughs collide when they have moved $kT/(\alpha h)$ metres. ■

The picture below shows the two paths in the (y, t) plane for $h = k = \alpha = 1$ and $T = 2$; the time gap $g(y)$ between the curves shrinks to zero at $y = 2$.



Checks

- Units: α has units m^{-1} (since $k/(\alpha z)$ is a speed), so αx is dimensionless, h/k is a time, and $kT/(\alpha h)$ is a length.
- If $T = 0$ the ploughs start together and the formula gives collision at distance 0; a bigger head start T pushes the collision further down the road, as it should.
- Numerically, for a range of random values of h, k, α and T, stepping each plough's motion forward in time (for the second, $\mathrm{d}y/\mathrm{d}t = 1/\big(\alpha(t - t_1(y))\big)$ from $y = 0$ at $t = T$) agrees with the formulae to better than 10^{-8}. With $h = k = \alpha = 1$ and $T = 2$, as in the picture, both ploughs reach $y = 2$ at time $e^2 - 1 \approx 6.39\,\mathrm{s}$.

Common slips

- Using depth $h + kt$ for the second plough too. It is ploughing a road that was cleared at time $t_1(y)$, so the depth is $k(t - t_1(y))$.
- Forgetting the initial condition $t = T$ at $y = 0$ when checking the solution; a solution of the differential equation alone is not enough.
- Writing $\mathrm{d}y/\mathrm{d}t$ instead of $\mathrm{d}t/\mathrm{d}y$: with time as the dependent variable the equations become linear and easy.

Beyond STEP

This is the classic "snowplough problem" family, and the trick of swapping the roles of dependent and independent variable to make an equation linear is used constantly in first-year differential equations. The uniqueness argument above is a baby case of the existence and uniqueness theorem for initial value problems. Notice too that the second plough's speed blows up as the snow in front of it thins to nothing, which is a sign of the model's limits rather than of reality. Boyce and DiPrima, *Elementary Differential Equations*, covers linear first-order equations, integrating factors and uniqueness.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.